Experiment 3. Case study on use of Neural Network in MATLAB for engineering problems

Aim

To forecast demand using Artificial Neural Network (ANN) in MATLAB.

Procedure

1. Collect historical demand data
2. Pre-process data (normalization)
3. Create ANN model in MATLAB
4. Train the network
5. Test the model
6. Predict future demand

Theory

Demand Forecasting is the process of predicting future demand based on historical data.

An **Artificial Neural Network (ANN)** is used because:

- It can learn patterns from past data
- It handles nonlinear relationships
- It gives accurate predictions

Structure of ANN:

- Input Layer → Past demand data
- Hidden Layer → Processing
- Output Layer → Predicted demand

Case Study (Engineering Example)

In industries or automation systems:

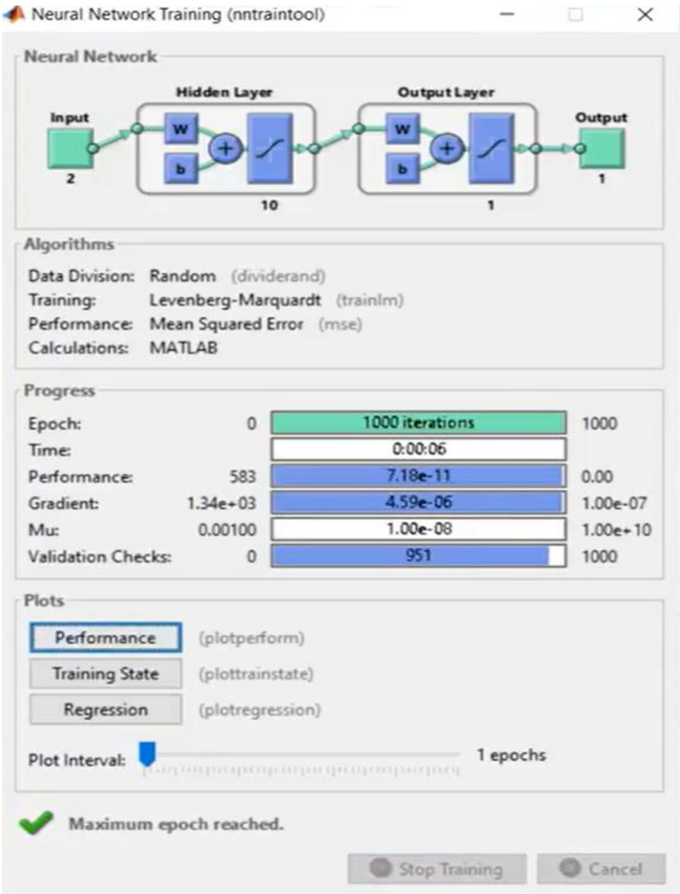
- Past sales/demand data is used
- ANN is trained on this data
- It predicts future demand

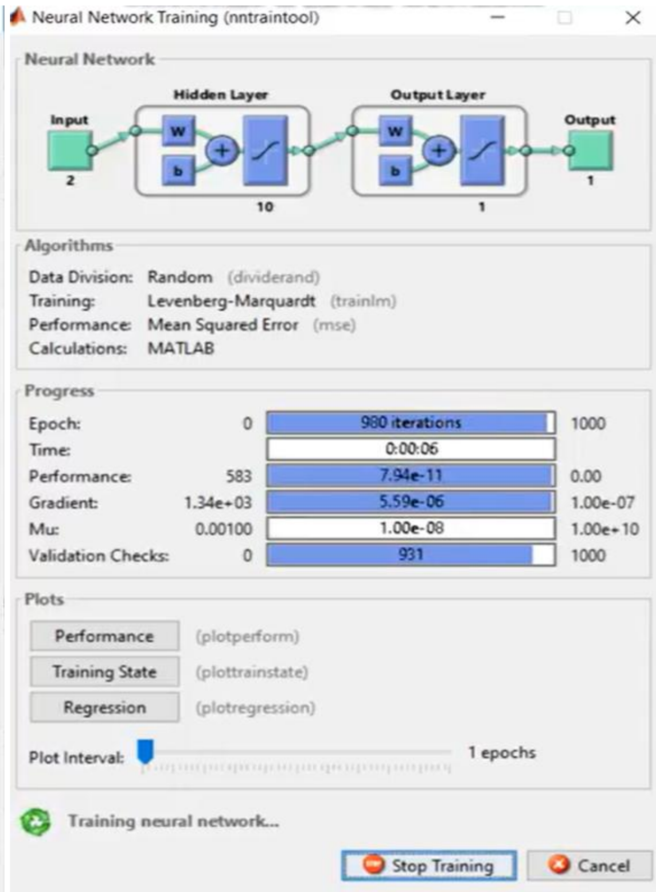
Example:

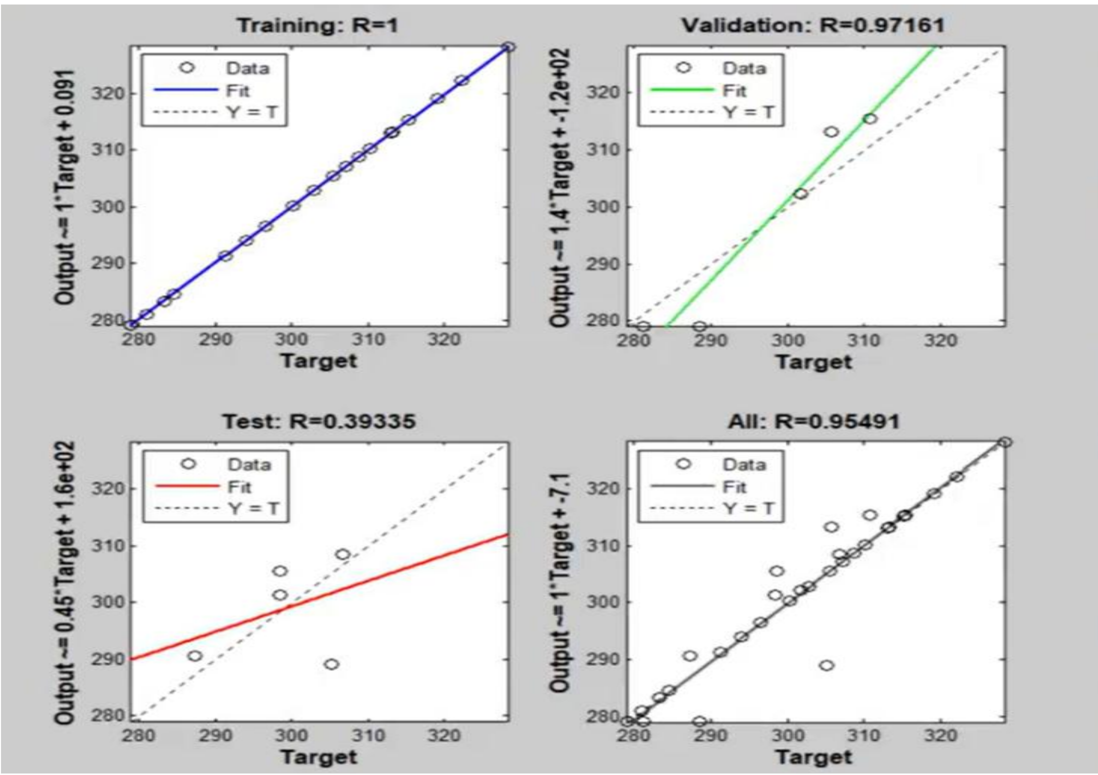
- Input: Demand of previous days
- Output: Demand of next day

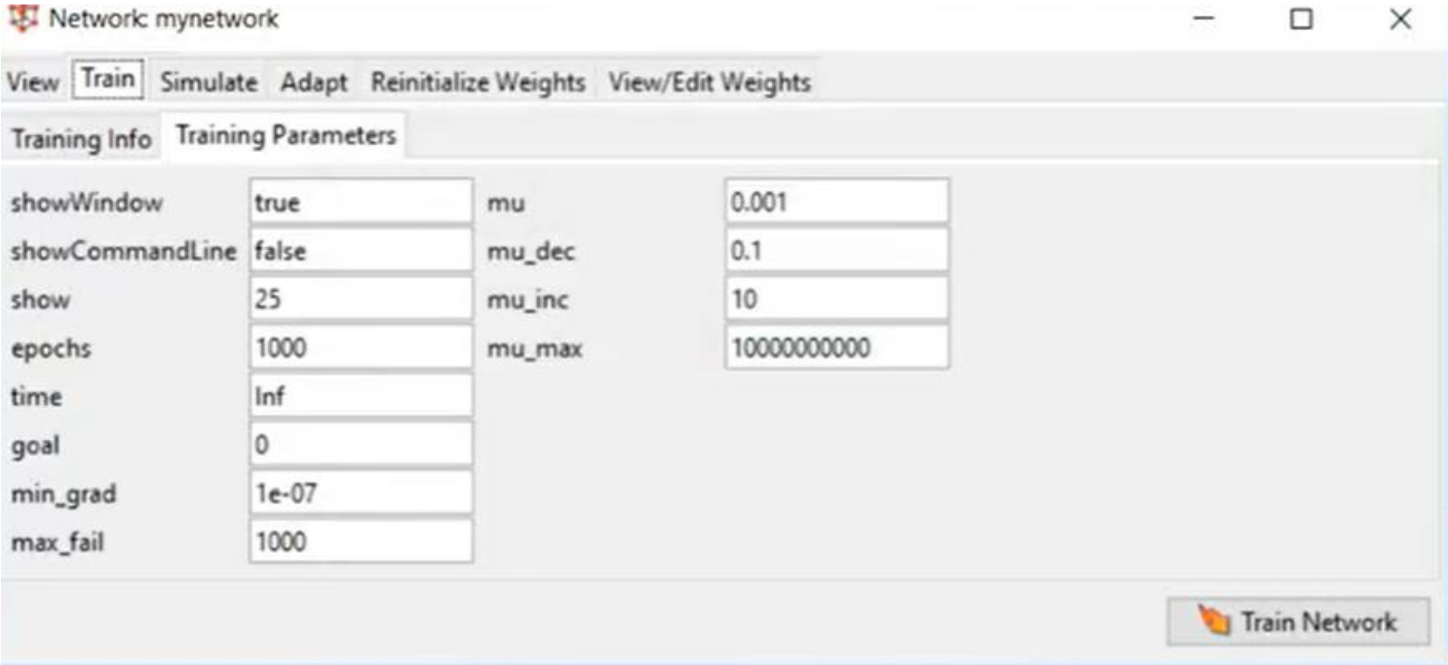
Results

- ANN predicts future demand based on past values
- Prediction improves with more training data
- Error reduces after training









Conclusion:

ANN is an effective tool for demand forecasting in engineering systems. It helps industries plan production and manage resources efficiently.